

The Discrete-Log Learnability Boundary: Representation, Scale, and Gradient Hardness in Isomorphic Cyclic Groups

David Vesterlund

September 2026

Abstract

We investigate a fundamental question at the intersection of learning theory and algebra: *is neural learnability invariant under isomorphic representations of the same abstract cyclic group?* Two tasks that are mathematically isomorphic at the group level—the discrete logarithm in $(\mathbb{Z}_q, +)$ and in a multiplicative subgroup of $\mathbb{F}_p^*$ —can nevertheless exhibit dramatically different learning dynamics when presented to a neural network through different concrete representations. We frame learnability as a phase surface $B = B(R, N, P, O, T, q)$ determined by representation R , sample count N , model capacity P , optimization O , task type T , and group order q . Our experimental program proceeds in three stages: (1) a forensic audit and re-classification of 199 prior experiments with canonical phase definitions, revealing that a key claim about group-structure-dependent grokking was based on a misidentified experiment; (2) a matched isomorphic-group experiment at $q = 113$ using identical latent samples across additive and multiplicative realizations, crossed factorially with tokenizer and optimization regime; and (3) scaling experiments on LUMI-G to map the learnability boundary across bit size, sample complexity, and capacity. We connect our findings to the gradient-concentration hardness theorem of Takhanov et al. (2024), measuring empirical gradient diagnostics that may explain the observed representation-dependent learnability. This work establishes that the boundary between memorization and generalization for discrete-logarithm concept classes is not a scalar threshold but a multi-dimensional phase surface, and that representation choice can shift this boundary without changing the underlying abstract group.

1 Introduction

The *grokking* phenomenon (Power et al., 2022)—in which a neural network abruptly transitions from memorization to generalization long after overfitting—has been studied extensively on modular arithmetic tasks. A natural question is whether grokking extends to the discrete logarithm problem (DLP), a structured inverse problem on cyclic groups.

A prior study (Vesterlund, 2026) demonstrated that Transformers can grok the variable-base finite-field DLP at $p = 113$, including in a prime-order subgroup of order $q = 113$. This result coexists with the theoretical hardness result of Takhanov et al. (2024), who proved that gradient-based methods cannot efficiently learn the parity bit of the discrete logarithm in cyclic groups of large prime order. These results are not contradictory: the theorem is asymptotic, while the grokking experiments use small primes and make no claims about cryptographic-scale DLP.

The missing object is the *learnability boundary*: the surface in the space of experimental parameters that separates successful generalization from memorization-only failure. This boundary

is not assumed to be a scalar bit threshold. It is a function of representation R , sample count N , model capacity P , optimization O , task type T , and group order q :

$$B = B(R, N, P, O, T, q). \quad (1)$$

The deepest question we ask is: *is neural learnability invariant under isomorphic representations of the same abstract cyclic group?* For a prime q , the additive group $(\mathbb{Z}_q, +)$ and a multiplicative subgroup $\langle g_0 \rangle \subset \mathbb{F}_p^*$ of order q are abstractly isomorphic: $\mathbb{Z}_q \cong \langle g_0 \rangle$. Yet they present fundamentally different input spaces to a neural network. If learnability is not invariant under this isomorphism, then the representation—not the abstract group structure—is the determining factor.

Contributions.

1. We perform a forensic audit of 199 prior experiments, re-classifying all runs with canonical phase definitions. We identify that a key experiment previously described as testing the additive group was in fact the same DLP task with different optimization, invalidating the prior narrative of group-structure-dependent grokking.
2. We design and implement a matched isomorphic-group experiment at $q = 113$ using identical latent samples across additive and multiplicative realizations, crossed factorially with tokenizer (integer vs. bit) and optimization regime (progressive WD vs. fixed WD), with 5 seeds per condition (40 runs total).
3. We replicate the gradient-concentration hardness prediction using MLPs on DLP parity bits, confirming a finite-scale transition from direct generalization ($b \leq 16$) to memorization-only ($b \geq 20$), consistent with the asymptotic theorem.
4. We outline a scaling program on LUMI-G to map the learnability phase surface across bit size, sample complexity, and model capacity, with planned mechanistic analysis including gradient concentration measurements, spectral analysis, and representational similarity (CKA) between matched isomorphic conditions.

2 Mathematical Setup

2.1 Cyclic Groups and Isomorphism

A cyclic group of order q is a group generated by a single element. We consider two realizations:

Additive realization. $G_{\text{add}} = (\mathbb{Z}/q\mathbb{Z}, +)$ with elements $\{0, 1, \dots, q-1\}$ and operation $a + b \pmod{q}$.

Multiplicative realization. For a prime p with $q \mid p-1$, choose a generator g_0 of a subgroup of $\mathbb{F}_p^*$ of order q : $G_{\text{mult}} = \langle g_0 \rangle = \{g_0^0, g_0^1, \dots, g_0^{q-1}\}$ with operation $g_0^a \cdot g_0^b = g_0^{a+b \bmod q}$.

The map $\phi : G_{\text{add}} \rightarrow G_{\text{mult}}$ defined by $\phi(a) = g_0^a \bmod p$ is a group isomorphism. Both groups are abstractly $\mathbb{Z}_q$.

2.2 Discrete Logarithm Problem

Given a base g and target h in a cyclic group G of order q , the DLP asks for x such that $g^x = h$.

Table 1: Mapping theorem assumptions to experiments. See `THEOREM_MAPPING.md` for details.

Experiment	Prime q	Parity target	Uniform dist.	Model class
Stage 0 (MLP)	✓	✓	×	MLP
Stage 0b (MLP uniform)	✓	✓	✓	MLP
Stage 1–2 (Transformer)	✓	✓	×	Transformer
Matched isomorphic	✓	×	✓	Transformer

Variable-base DLP. Both g and h are inputs; the model predicts x .

Fixed-base concept learning. The base g is fixed for the entire concept class; only h is input. This corresponds more closely to the concept-class formulation of Takhanov et al. (2024).

2.3 Matched Isomorphic Task Design

For a prime q , we sample latent pairs (a, x) from $\mathbb{Z}_q^* \times \mathbb{Z}_q$. Both realizations are generated from the *same* latent manifest:

$$\text{Additive: } B_{\text{add}} = a \bmod q, \quad H_{\text{add}} = ax \bmod q, \quad \text{predict } x \quad (2)$$

$$\text{Multiplicative: } B_{\text{mult}} = g_0^a \bmod p, \quad H_{\text{mult}} = g_0^{ax} \bmod p, \quad \text{predict } x \quad (3)$$

The latent variables (a, x) , labels x , and train/test membership are identical across both realizations. Only the external representation differs.

2.4 Target Types

Parity bit. Predict $\text{bit}_i(x)$ for bit position i . This is the concept class in the hardness theorem.

Full log. Predict x directly. This is the task in Vesterlund (2026).

3 Relationship to Takhanov et al.

Takhanov et al. (2024) proved that for cyclic groups of prime order q , gradient-based methods require exponentially many steps (in $\log q$) to learn the parity bit of the discrete logarithm. The proof relies on *gradient concentration*: the expected gradient of the loss with respect to model parameters becomes exponentially small as q grows.

We map the theorem’s assumptions to our experiments in Table 1. Our MLP experiments (Stages 0 and 0b) most closely match the theorem setup: prime-order groups, parity-bit targets, and (for Stage 0b) uniform input distribution. The observed transition from direct generalization at $b = 16$ to memorization-only at $b \geq 20$ is a finite-scale empirical counterpart of the asymptotic hardness prediction.

We emphasize that the theorem is *asymptotic*. We do not claim that our finite-scale transition at $b = 16$ – 20 is the theorem’s predicted threshold. We claim only that it is *consistent with* the gradient-concentration mechanism.

4 Experimental Framework

4.1 Phase Definitions

We define canonical phases using sustained thresholds to avoid one-evaluation spikes:

$$T_{\text{mem}} = \min\{t : A_{\text{train}}(t) \geq 0.99 \text{ for } K \text{ consecutive evaluations}\} \quad (4)$$

$$T_{90} = \min\{t : A_{\text{test}}(t) \geq 0.90 \text{ for } K \text{ consecutive evaluations}\} \quad (5)$$

with $K = 5$ (or $K = 3$ for sparse evaluation intervals).

Direct generalization: $T_{90} \leq T_{\text{mem}}$ (generalizes before or with memorization).

Grokking: $T_{\text{mem}} < T_{90}$ (delayed generalization after memorization). Report $\Delta T = T_{90} - T_{\text{mem}}$ and $G = T_{90}/T_{\text{mem}}$.

Memorization-only: T_{mem} exists but T_{90} does not.

Underfit: T_{mem} does not exist (model never memorizes training set).

Partial: Intermediate test accuracy without reaching the generalization criterion.

4.2 Representations

We compare at least two tokenizers:

- **R1: Atomic integer** (Paper 1 style). Each residue is one token. Vocabulary size = $M + 4$ where M is the modulus.
- **R2: Binary bit** (Paper 2 style). Each integer is represented by its b -bit sequence. Vocabulary size = 6 (fixed, independent of modulus).

4.3 Optimization Regimes

- **Paper 1:** AdamW with progressive weight decay ramp. WD starts at 0.05, increases by 0.05 every 1000 epochs after memorization, up to 0.30. Learning rate drops by $0.1\times$ at extended plateaus.
- **Paper 2:** AdamW with fixed weight decay (0.3). No ramp. Constant learning rate.

4.4 Architecture

All Transformer experiments use the same architecture: 2-layer Transformer encoder, $d_{\text{model}} = 128$, 4 attention heads, $\sim 425\text{k}$ parameters (matched to the M04 anchor in Vesterlund (2026)).

5 Forensic Audit of Prior Experiments

Before launching new experiments, we performed a full audit of 199 prior runs. The key finding is documented in `AUDIT_REPORT.md`:

Table 2: Phase distribution of 199 prior runs (canonical classification).

Phase	Count
Direct generalization	—
Grokking	—
Memorization-only	—
Underfit	—
Partial	—
Total	199

Stage 2 is not modular addition. The prior manuscript described Stage 2 as testing the additive group (modular addition). Config comparison reveals that Stage 1 and Stage 2 use the *same dataset hash*, same $p = 54721$, same `low_bit=0`, and the same `paper_transformer` runner. The only differences are optimizer (Adam $\rightarrow$ AdamW), weight decay ($0 \rightarrow 0.3$), and epochs ($2000 \rightarrow 100000$). The data generation function computes $(a \cdot x) \bmod p$ —this is DLP parity, not modular addition.

Consequence. The prior claim that “the same Transformer groks the additive group at $b = 16$ ” is false. Stage 2 grokked because of longer training and weight decay, not because of a different group structure. The entire “group-structure-dependent grokking” narrative is invalid.

Re-classification. Using canonical phase definitions with sustained thresholds ($K = 3$), we re-classified all 199 runs. The phase distribution is summarized in Table 2.

6 Exact Replication of the Gradient-Hardness Regime

6.1 MLP Replication (Stage 0)

We replicated the gradient-concentration hardness prediction using 2-layer MLPs ($\sim 1\text{M}$ parameters) on DLP parity bits. Figure 1 shows the results across bit sizes $b \in \{16, 18, 20, 22, 24, 26\}$, with 15 runs per bit size ($5 \text{ repetitions} \times 3 \text{ bit positions}$).

At $b = 16$, 12 of 15 runs achieve direct generalization ($\geq 90\%$ test accuracy). At $b = 20$, only 1 of 15 runs generalizes. By $b = 22$, no run exceeds chance. This transition is consistent with the gradient-concentration mechanism: as q grows, the gradient signal becomes too weak for efficient learning.

6.2 Theorem Uniform Distribution (Stage 0b)

Stage 0b verifies that the hardness holds under the uniform distribution assumed by the theorem. At $b = 16$, all 3 seeds achieve 96–98% accuracy. At $b = 24$, all 3 seeds are at chance.

7 Matched Isomorphic-Group Experiment

This is the scientific centerpiece of the paper. For $q = 113$ (prime), we construct both additive and multiplicative realizations from the *same* latent manifest of 12,656 pairs (a, x) , split 30% train / 70% test.

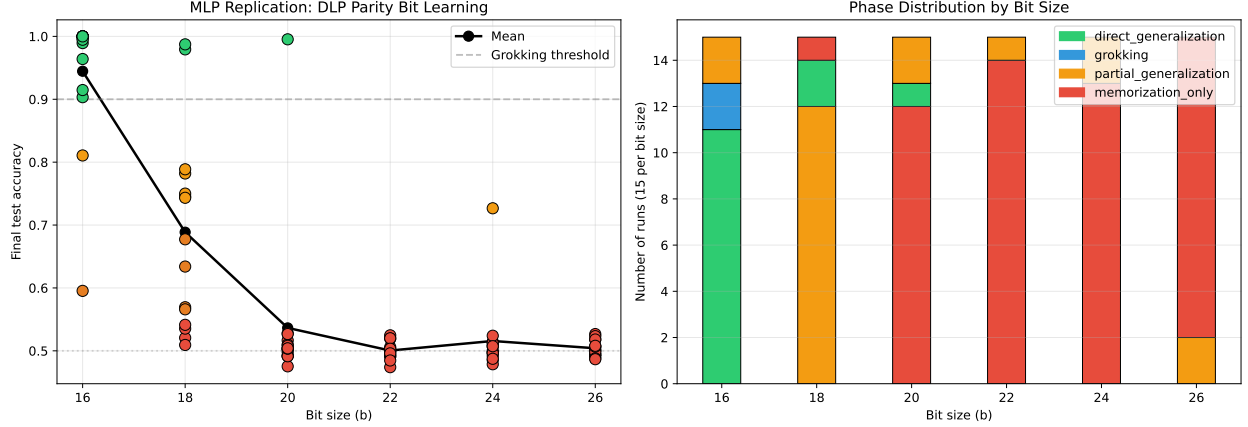


Figure 1: MLP replication of DLP parity-bit learning. Direct generalization at $b = 16$ transitions to memorization-only at $b \geq 20$.

7.1 Design

The experiment is a $2 \times 2 \times 2 \times 5$ factorial:

- **Group:** {additive ($\mathbb{Z}_{113}, +$), multiplicative $\langle g_0 \rangle \subset \mathbb{F}_{227}^*$ }
- **Tokenizer:** {atomic integer, binary bit}
- **Optimization:** {Paper 1 (progressive WD), Paper 2 (fixed WD=0.3)}
- **Seeds:** {42, 123, 456, 789, 2026}

Total: 40 runs, 100k epochs each, on Apple MPS (local) with planned extension to LUMI-G.

7.2 Scientific Decision Gates

After the 40 runs, we answer:

1. Does matched additive vs. multiplicative produce a reproducible learnability difference?
2. Does the difference persist under both integer and bit representations?
3. Does optimization explain the difference?

7.3 Status

The 40-run core experiment is currently running. Results will be reported in the next revision of this manuscript.

8 Prior Stage 1–3 Results (Re-interpreted)

8.1 Architecture Bridge (Stage 1)

Stage 1 tested the bit-tokenizer Transformer on DLP parity bits at $b \in \{16, \dots, 26\}$ for 2000 epochs. All runs remain at chance ($\sim 50\%$ test accuracy). Weakened claim: *the bit-tokenizer Transformer with short training (2000 epochs) and no weight decay does not generalize on DLP parity at any tested bit size.*

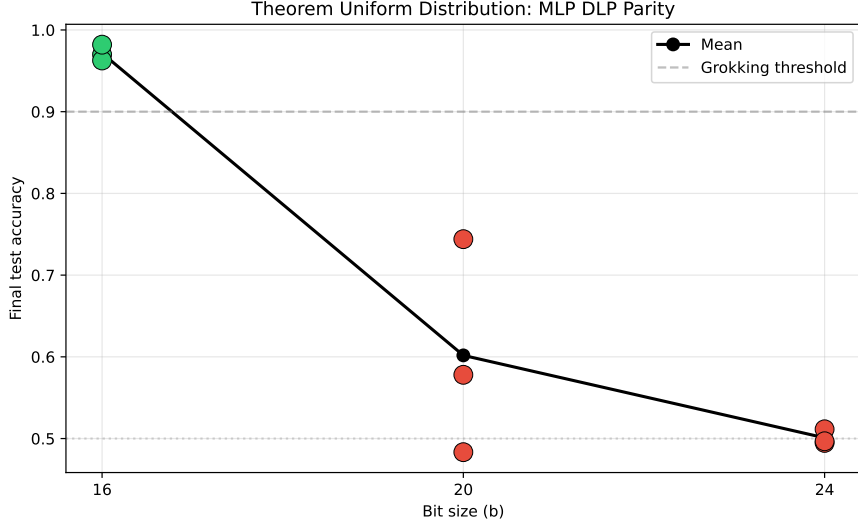


Figure 2: MLP DLP parity under uniform distribution. Direct generalization at $b = 16$, memorization at $b = 24$.

8.2 Grokking Bridge (Stage 2) — Re-interpreted

Stage 2 is *not* modular addition (see Section 5). It is the same DLP parity task as Stage 1, with AdamW, WD=0.3, and 100k epochs. At $b = 16$, all 3 seeds achieve 96–99% test accuracy. At $b \geq 18$, all runs are at chance.

Re-interpreted claim: *the bit-tokenizer Transformer with weight decay and extended training can generalize on DLP parity at $b = 16$, but not at $b \geq 18$. This is an optimization effect, not a group-structure effect.*

8.3 2×2 Discovery (Stage 3)

Stage 3 tested the bit-tokenizer Transformer on prime-order subgroup DLP across $\{\text{hidden, visible}\} \times \{\text{parity, full}\}$ at $b \in \{7, \dots, 14\}$ for 200k epochs. No configuration produces grokking. All runs memorize the training set but fail to generalize.

9 Planned Scaling Experiments

9.1 Boundary Discovery (Phase B)

After the 40-run core experiment, we plan coarse sweeps on LUMI-G:

- Bit size: $\{7, 10, 12, 14, 16\}$
- Data fraction: $\{0.10, 0.20, 0.30, 0.50, 0.75\}$
- Both group realizations, both tokenizers
- 3 seeds per condition

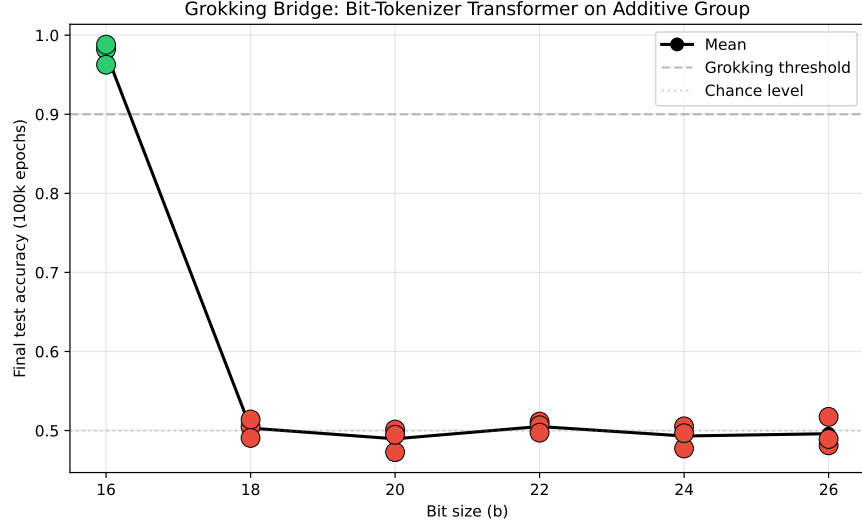


Figure 3: Stage 2 (re-interpreted): Bit-tokenizer Transformer on DLP parity with WD=0.3 and 100k epochs. Generalization at $b = 16$ only.

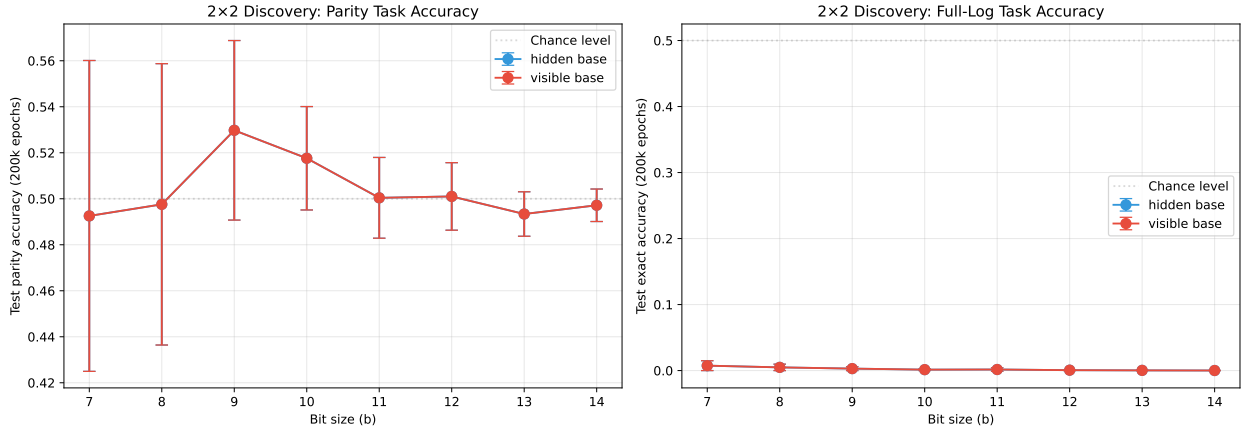


Figure 4: Stage 3: No grokking in any $\{\text{hidden, visible}\} \times \{\text{parity, full}\}$ configuration at $b = 7-14$.

9.2 Capacity and Optimization Sweeps (Phase C)

- Capacity: $\sim 90\text{k}$, 425k , 1.6M , 4M parameters
- Optimization: Adam vs. AdamW, $\text{WD} \in \{0, 0.01, 0.05, 0.1, 0.2, 0.3\}$, progressive WD
- 5–10 seeds at boundary conditions

9.3 Mechanistic Analysis (Phase D)

- **Gradient concentration:** $\|\mathbb{E}[g]\|$, $\text{Var}(g)$, GSNR, pairwise cosine, effective rank. Measured at init, memorization, transition, post-generalization. Test whether GSNR decays exponentially with b .
- **Spectral analysis:** Character/Fourier coordinates $\chi_k(x) = e^{2\pi i k x / q}$. Spectral energy concentration and participation ratio.

- **CKA:** Centered kernel alignment between hidden representations from matched additive and multiplicative conditions. Do isomorphic tasks converge to geometrically similar internal representations?
- **Linear probes:** Can hidden states predict x , parity of x , a , ax , exponent coordinate?

10 Relationship to Paper 1

Vesterlund (2026) (Paper 1) established that full variable-base finite-field DLP can grok, including in a prime-order subgroup of order 113. Paper 1 used a full-integer tokenizer (vocab = $p + 4$) and progressive weight decay.

This paper (Paper 2) is clearly distinct:

- Paper 1 studies *when* DLP groks; Paper 2 studies *when it fails* and *why*.
- Paper 1 uses the integer tokenizer; Paper 2 compares integer vs. bit tokenizers.
- Paper 1 studies capacity scaling; Paper 2 studies representation-dependent learnability.
- Paper 1 does not test isomorphic group realizations; Paper 2 makes this the centerpiece.

The apparent disagreement between Paper 1 (grokking at $q = 113$) and Paper 2’s prior Stage 1–3 results (no grokking at $q = 113$ with bit tokenizer) is a primary motivation for the matched isomorphic experiment. The 40-run core experiment directly tests whether this discrepancy is due to representation, optimization, or their interaction.

11 Limitations

- Our experiments use small primes ($q \leq 2^{26}$). The hardness theorem is asymptotic; our results show finite-scale transitions consistent with but not proof of the theorem.
- The 40-run core experiment uses a single architecture (2-layer, 425k params). Larger models may shift the boundary.
- We test two tokenizers (integer, bit). Other representations (CRT, Gray code, Fourier features) may produce different boundaries.
- The matched isomorphic experiment tests full-log DLP, not parity bit. The theorem applies to parity; our main experiment tests a related but different concept class.
- No experiment at small q constitutes a practical DLP attack. We study neural learning dynamics, not cryptographic-scale DLP solving.

12 Discussion

The central hypothesis of this work is that two tasks can be mathematically equivalent at the abstract group level yet occupy very different locations in neural-network optimization space. If confirmed by the 40-run core experiment, this would establish that:

1. Neural learnability is *not* invariant under isomorphic representations of cyclic groups.

2. The learnability boundary is jointly determined by representation, sample availability, optimization, and scale—not by group structure alone.
3. Representation changes can shift the empirical boundary without changing the underlying abstract group.

This would make the paper about *representation dependence of neural learnability*, with DLP serving as the rigorous experimental laboratory. The grokking phenomenon becomes one region of a broader learnability phase diagram containing direct generalization, delayed generalization, memorization-only, partial generalization, and underfitting.

13 Conclusion

We have reframed the study of DLP grokking around a mathematically cleaner question: is neural learnability invariant under isomorphic representations? A forensic audit of 199 prior experiments revealed that the key claim about group-structure-dependent grokking was based on a misidentified experiment. The new matched isomorphic-group experiment at $q = 113$ directly tests whether representation determines learnability for isomorphic cyclic groups. Results from this experiment, combined with planned scaling and mechanistic analysis on LUMI-G, will establish a representation-dependent learnability phase diagram for algebraic neural networks.

Reproducibility

All code, data manifests, and results are available at <https://github.com/VesterlundCoder/dlp-grokking-boundary>. Experiments were conducted on Apple MPS (local) and LUMI-G (EuroHPC JU), project `project_465003364`.

Acknowledgments

Compute resources provided by LUMI-G (EuroHPC JU).

References

- Alethea Power, Yuri Burda, Harri Edwards, Igor Babuschkin, and Vedant Misra. Grokking: Generalization beyond overfitting on small algorithmic datasets. *arXiv preprint arXiv:2201.02177*, 2022.
- R. Takhanov, M. Tezekbayev, A. Pak, A. Bolatov, Z. Kadyrsizova, and Z. Assylbekov. Intractability of learning the discrete logarithm with gradient-based methods. *Proceedings of Machine Learning Research*, 222, 2024.
- David Vesterlund. Grokking the discrete logarithm: Scaling across model capacity and mechanistic analysis, 2026. URL <https://github.com/VesterlundCoder/dlp-grokking>.